MICROSAR vLinkGen

Technical Reference

Compiler Specific Linker Script Generator

Version 2.2.1

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Document Information

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Reference Documents

No.	Source	Title	Version
[1]	AUTOSAR	General Specification of Basic Software Modules	R4.2.1



Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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1 Introduction

This document describes the functionality, API and configuration of the MICROSAR BSW module vLinkGen.

Supported Configuration Variants:	pre-compile	
Vendor ID:	vLinkGen_VENDOR_ID	30 decimal (= Vector-Informatik, according to HIS)
Module ID:	vLinkGen_MODULE_ID	255 decimal (according to ref. [1])

The vLinkGen (Vector linker script generator) module shall provide an abstract way to create linker scripts, so that code, constants and variables (linker symbols) in a binary are placed at the intended location in the target memory (e.g. ROM, RAM, etc.).

The basic idea of vLinkGen is to have a compiler independent description (except for assignment to hardware specific memory areas) of the linker script to be able to apply it in a similar way for several projects.

The configuration of the linker script can be done completely without knowledge of the compiler specific linker script language. The vLinkGen then generates the linker script in the appropriate format for the selected compiler.

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1.1 Architecture Overview

The following figure shows where the vLinkGen is located in the AUTOSAR architecture.

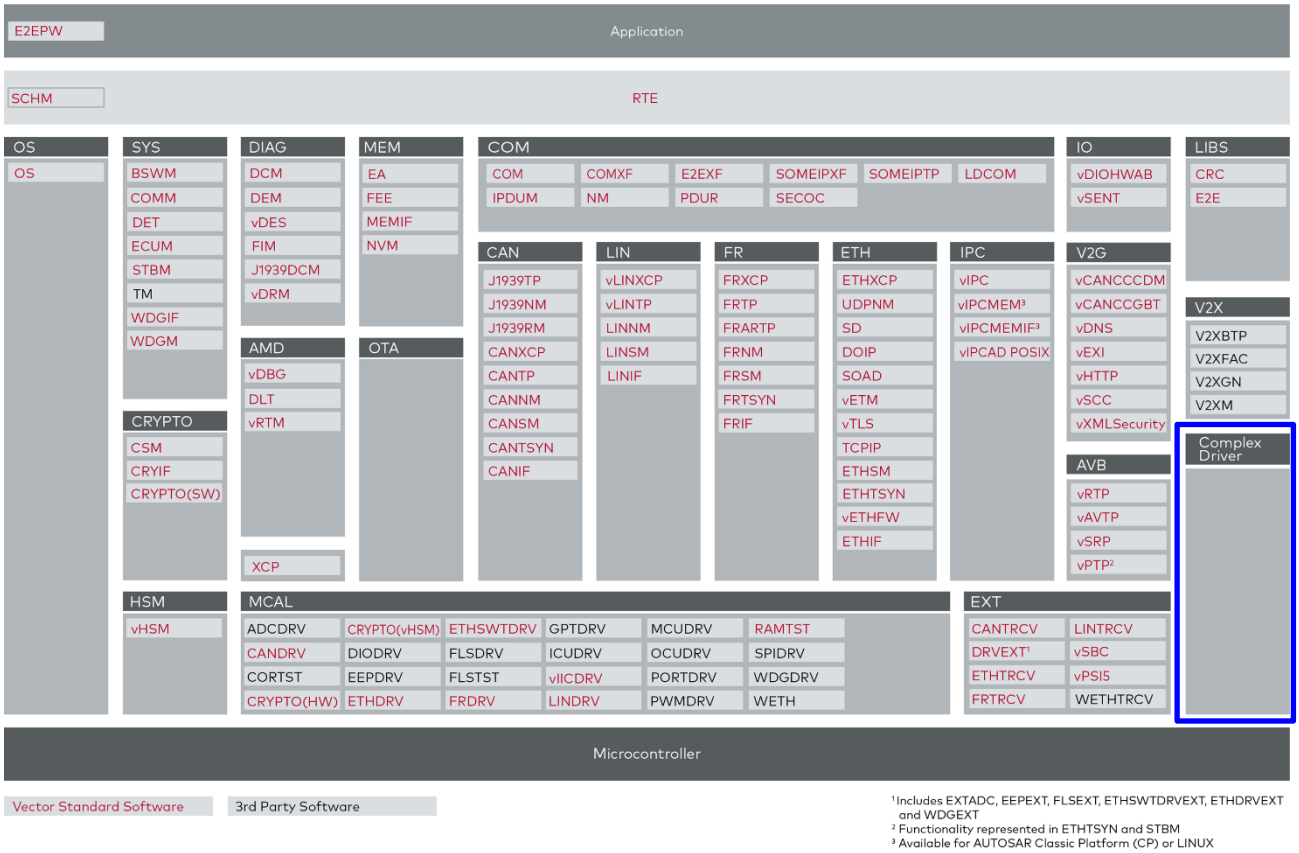


Figure 1-1 AUTOSAR 4.3 Architecture Overview

1.2 Supported Compilers

The following table shows all supported compilers.

Compiler Name	Version Restrictions
ARM	only 5.x.x, 6.x.x or later
Wind River Diab	-
GCC	-
Green Hills	-
HighTec (GCC)	-
HighTec (LLVM)	-
IAR	only 7.x.x or later
Linaro	-
Renesas	-
Tasking	-
Texas Instruments	-

Table 1-1 Supported Compilers



Note
Not all versions were tested for each compiler. It is assumed that all common versions will work with the generated linker scripts.

1.3 Definitions of Terms

This chapter describes the different terms in the vLinkGen and shows a schematic configuration example in Figure 1-2.

- **Hardware Memory Areas** (`vBaseEnvMemLayoutHwRegion`) are provided by vBaseEnv module. They represent the actual memory layout on the hardware and cannot be changed.
- **Memory Regions** (`vLinkGenMemoryRegion`), which can be assigned to the hardware memory areas, are a logical abstraction layer to separate the configuration from the actual memory layout and to make it hardware independent. See `Code_Flash` and `Data_Flash`, mapped to `FLASH` in Figure 1-2.
- Memory Regions can be divided into **Memory Region Blocks** (`vLinkGenMemoryRegionBlock`), which can be linked to the lower or upper boundary of the region. They are used for the actual memory definitions in the generated linker script. See `Code_Flash_0` and `Code_Flash_1`, mapped to `CodeFlash` in Figure 1-2.

- **Logical Groups** (`vLinkGenLogical*Group`) are a logical abstraction layer to group multiple Section Groups or Linker Symbols and map them to a Memory Region Block. See `Appl_Const`, mapped to `Data_Flash_0` in Figure 1-2.
- **Section Groups** (`vLinkGen*SectionGroup`) are used to group multiple Linker Sections of a kind together. In compiler manuals they are often called ‘Output Sections’. See `Code_Default` containing Linker Sections and is mapped to `Code_Flash_0` in Figure 1-2.

  The size of these groups depends on the concrete content of the Linker Sections. Also, their start address depends on the configured position and the size of previous groups, which are mapped to the same Memory Region Block.
- **Linker Sections** are the smallest entity in the vLinkGen and contain the concrete content of the different linker symbols (functions, constants, variables, etc.) assigned to them in the source code. In compiler manuals they are often called ‘Input Sections'.



Caution
The following figure shows only a schematic example of a configuration and is by no means to be regarded as a template or complete. It serves only as an illustration of the existing types and their relationships to each other.

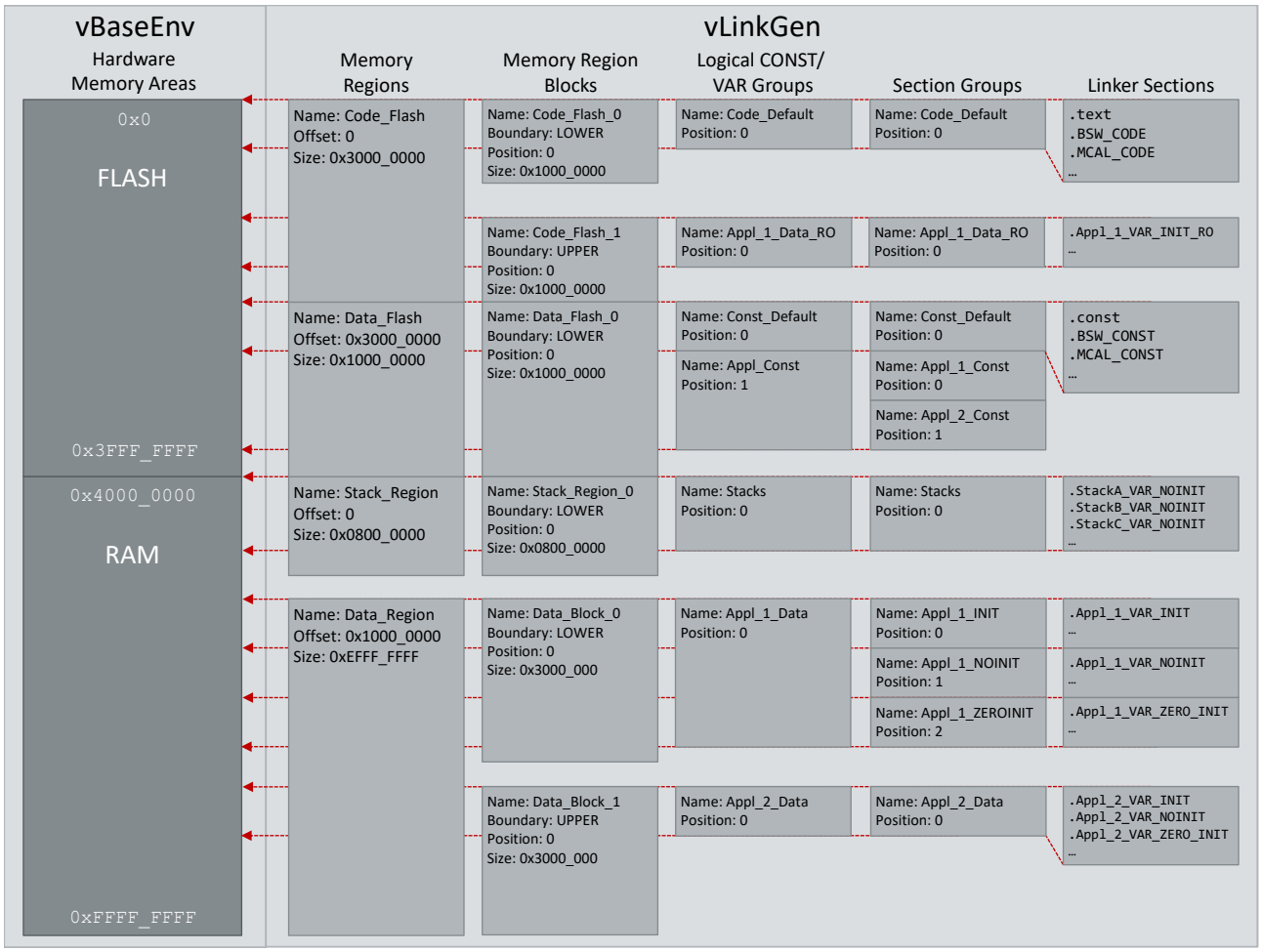


Figure 1-2 Schematic Configuration Example

2 Functional Description

The following chapters contain some detail descriptions of the vLinkGen features.



Caution
Please consider that not all features are supported by all compilers. If a feature is not supported, the corresponding parameters are either set to read-only or an error or warning message occurs when they are activated in the DaVinci Configurator.

2.1 File Generation

The vLinkGen provides the possibility to generate the linker script in different variants with different content. These can be selected via the **File Generation** (`vLinkGen/vLinkGenGeneral/vLinkGenFileGeneration`) parameter in the configuration.

For details about the different file types and their content, please see Table 2-1 below.

The names with which the files are generated can be found in chapter 3.1.

Details about variant handling in vLinkGen can be found in chapter 2.2.

STANDARD	Generates a complete compiler specific linker script. ▶ Logical Groups that are assigned to one or more variants are encapsulated with C preprocessor macros. That means that if variants are configured, this file may need to be preprocessed before it can be passed to the linker. This applies to all linkers that do not have their own preprocessor that can resolve C preprocessor macros (e.g. `#if defined (VariantName)`). Also the variant must be known during preprocessing. This means it must be passed to the preprocessor (e.g. via options with `-DVariantName`). ▶ Logical Groups that are not assigned to a variant are generated without macros and are therefore always active.
ONE_FILE_PER_VARIANT	Generates a complete compiler specific linker script per variant. ▶ These files can be used as they are and passed to the linker via specific option. Preprocessing of these files is not necessary. ▶ All Logical Groups that are assigned to one or more variants are generated in the corresponding variant specific files. ▶ Logical Groups that are not assigned to a variant are generated in all files and are therefore always active.
MODULE_SNIPPETS	Generates several linker script snippets containing all Logical Groups (including all Section Groups) with a module reference. ▶ Please consider the notes on snippets in chapter 2.1.1.
VARIANT_SNIPPETS	Generates several linker script snippets containing all Logical Groups (including all Section Groups) of a variant. ▶ Please consider the notes on snippets in chapter 2.1.1.

LOGICAL_GROUP_SNIPPETS	Generate several linker script snippets containing all Section Groups of a Logical Group. ▶ Please consider the notes on snippets in chapter 2.1.1.

Table 2-1 File Generation Types

2.1.1 Notes on Snippets

- Snippets are not supported for all compilers.
- Snippets only contain the corresponding Logical Groups and nothing else.
- Snippets cannot be passed directly to the linker. They can only be included by other linker scripts.
  - However, the names of the Memory Region Blocks (e.g. FLASH) from the vLinkGen configuration must match the memory definitions from the linker script into which the snippets are to be included.
- If variants are configured, Logical Groups inside the snippets are encapsulated the same way as for STANDARD generation.
- Snippets never contain memory definitions and global (PRE or POST) user groups (See 2.3).

2.2 Variant Handling

Sometimes it can be necessary to compile a project in several build processes with different memory layouts. Therefore, vLinkGen provides the possibility to configure several variants of a linker script in a single project and then generate them separately. This means that several linker scripts with different memory layouts can be generated from one configuration. These files can then be used for the several build processes.



Caution
If variants are configured, exactly one of them must always be defined in a build process. For example, by passing it as compiler and/or linker option (e.g. `-DVariantName`).

Variants are configured globally, and then referenced by one or more Logical Groups (`vLinkGenLogical*Group`). The following points must be observed:

- A Logical Group can also be assigned to several variants. If it is, it is therefore valid in each of them.
- Logical Groups that are not assigned to a variant are to be understood as global. This means they are active in every variant.

Depending on the configured variants and the settings of the **File Generation** (see chapter 2.1), the corresponding linker scripts are then generated.

An example how variants are configured can be found in chapter 5.2.6.



Example
The following example shows a Logical Group (including one Section Group) which is assigned to two variants (Variant_A and Variant_B) and the **File Generation** is set to `STANDARD`.

```
#if defined (Variant_A) || defined (Variant_B)
  .MySectionGroup :
  {
    _MySectionGroup_START = ABSOLUTE(.);
    *(*.My_Section)
    _MySectionGroup_END = ABSOLUTE(. - 1);
    _MySectionGroup_LIMIT = ABSOLUTE(.);
  } > RAM

  _MyLogicalGroup_ALL_START = _MySectionGroup_START;
  _MyLogicalGroup_ALL_END = _MySectionGroup_END;
  _MyLogicalGroup_ALL_LIMIT = _MySectionGroup_LIMIT;
#endif
```

2.2.1 Variant Specific Mappings

In addition to the normal variant handling, there is also the possibility to link certain Logical Groups (respectively all their contained Section Groups) to different Hardware Memory Areas in different variants.

What means a Logical Group, for example, can be linked to memory area `RAM_A` in Variant A and to totally different memory area `RAM_B` in Variant B. So, there is no need to duplicate the Logical Group and map it to different Memory Region Blocks. See example below.

The following behavior applies when setting the Memory Region Block references during generation:

- For all variants, the "Default Memory Region Block" references from the Logical Group are used by default
- If a variant is configured in a specific Variant mapping (`vLinkGen*GroupVariantSpecifics/vLinkGen*GroupSpecificVariant`), then the "Overwriting Memory Region Block" references are used
- Where the "overwriting" references in the variant specifics are optional
  - This means that for a variant, for example, only the ROM/RAM copy can be linked elsewhere



Note
Variant Specific Mappings also work with all **File Generation** types specified in chapter 2.1.



Example
The following example shows a Logical Group (including one Section Group) which is assigned to two variants (Variant_A and Variant_B) and the **File Generation** is set to `STANDARD`. In addition the default Memory Region Block (`RAM_A`) is specifically overwritten to `RAM_B` for Variant_B. So `MySectionGroup` will be linked to different memory areas in the variants.

```
#if defined (Variant_A) || defined (Variant_B)
  .MySectionGroup :
  {
    _MySectionGroup_START = ABSOLUTE(.);
    *(*.My_Section)
    _MySectionGroup_END = ABSOLUTE(. - 1);
    _MySectionGroup_LIMIT = ABSOLUTE(.);
#if defined (Variant_A)
  } > RAM_A
#endif
#if defined (Variant_B)
  } > RAM_B
#endif

  _MyLogicalGroup_ALL_START = _MySectionGroup_START;
  _MyLogicalGroup_ALL_END = _MySectionGroup_END;
  _MyLogicalGroup_ALL_LIMIT = _MySectionGroup_LIMIT;
#endif
```

2.3 User Code

The vLinkGen offers the possibility to generate compiler specific user code (user code group) at almost any position in the linker script. This allows the generated linker script to be extended with additional code snippets. This could be for example include of another linker script or generate additional comments.

The **Placement** (`vLinkGenUserGroupPlacement`) provides different options to choose from to generate the user code group to a specific location in the linker script. For details about the available options, see Table 2-2.

PRE	The user code is generated at the beginning of the linker script.
GROUP_LIST	The user code is generated within the Logical Groups list at the specified position. Note that this option is not available for all compilers.
POST	The user code is generated at the end of the linker script.

Table 2-2 User Group Placement Options

To configure such user code groups, create a `vLinkGenLogicalUserGroup` and select where it should be generated via the **Placement**. The `vLinkGenUserSectionGroup` contains the compiler specific code which is generated in the linker script exactly as configured.

2.4 Initialization Tables

Depending on the configuration, vLinkGen generates some structures (initialization tables) which can be used to:

- Initialize Memory Region Blocks with zeros (ECC initialization). See chapter 2.4.1.
- Initialize variables concrete values (INIT) or zeros (ZERO_INIT). See chapter 2.4.2.
- Copy code from ROM to RAM to be executed there. See chapter 2.4.3.

The initialization itself is usually done by the startup code of vBRS. This code takes over the writing of the memory areas with zeros or the copying of the values from ROM to RAM.

The following Table 2-3 shows the generated structures and their content.

The type definitions can be found in chapter 4.1.

Name	Content
`vLinkGen_MemAreaSet` **`vLinkGen_ZeroInit_<Stage>_BlocksSet`**	Memory Region Blocks to be initialized with zeros at a certain startup stage. Available stages can be found in Table 2-4.
`vLinkGen_MemAreaSet` **`vLinkGen_ZeroInit_<Stage>_GroupsSet`**	VAR Section Groups to be initialized with zeros at a certain startup stage. Available stages can be found in Table 2-4.
`vLinkGen_RamMemAreaSet` **`vLinkGen_Init_<Stage>_GroupsSet`**	Section Groups to be copied from ROM to RAM at a certain startup stage. This can be VAR Section Groups with initialization data or CONST Section Groups with code to be executed from RAM. Available stages can be found in Table 2-4.

Table 2-3 Initialization Tables

The following Table 2-4 shows all available initialization stages. The initialization stage determines the time of initialization during ECU startup.

It can be configured for each Memory Region Block, VAR or CONST Section Group separately via the **Init Stage** parameter. For Section Groups the value is only evaluated if the **Init Policy** is set to value other than `NONE`.

Consider that not all stages are available for Memory Region Blocks and CONST Section Groups.

Init Stage	Description
EARLY	Initialize always directly after reset. Shall only be used if really necessary. Since the PLL might not be initialized at that time, the initialization of large areas may take a long time.
ZERO	Always initialized first during the default vBRS startup routine.
ONE	Always initialized second during the default vBRS startup routine.By default, this value should be used if an initialization is required.
HARD_RESET_ONLY	The same as stage ONE. However, initialization is only done if a hardware reset occurred.
TWO, THREE	Optional init stages, which are initialized accordingly after the HARD_RESET_ONLY stage. Shall only be used if really necessary. One use case might be to reset the Wdg between the initialization of two stages.

Table 2-4 Available Init Stages

2.4.1 ECC Memory Initialization

On some platforms it is necessary to initialize certain memory areas (ECC protected memory areas) before the first read or write access. This is usually done by writing the complete area with zeros. The vLinkGen provides the possibility to enable this initialization for each Memory Region Block.

An example how to configure zero initialized Memory Region Blocks can be found in chapter 5.2.2.

2.4.2 Variable Initialization

The vLinkGen also offers the possibility to initialize variables on ECU startup with either predefined values or zeros. No initialization is of course also possible. However, the following points must be observed:

- Since the type of initialization is configured for a Section Group, it applies to all underlying Linker Sections. Which also means that within a Section Group there shall only be Linker Sections which need the same type of initialization.
- The type of initialization of a Section Group can only be INIT, ZERO_INIT or NONE at the same time. Multiple selection is not possible.

An example how to configure VAR Section Group to be initialized with predefined values or zero can be found in chapter 5.2.3.



Note
Some linkers provide the possibility to create own so-called "copy tables". These cannot be used together with vLinkGen and may have to be disabled via options.

2.4.3 RAM Code Execution

Like the VAR Section Groups, vLinkGen also offers the possibility to copy CONST Section Groups from ROM to RAM. This may be necessary if code is to be executed in RAM.

An example how to configure a CONST Section Group to be copied from ROM to RAM can be found in chapter 5.2.4.

2.5 Initialization

The vLinkGen does not provide any C initialization functions.

2.6 States

The vLinkGen does not provide any states.

2.7 Main Functions

The vLinkGen does not provide any main functions.

2.8 Error Handling

2.8.1 Development Error Reporting

The vLinkGen does not provide development error reporting.

2.8.2 Production Code Error Reporting

The vLinkGen does not provide production code error reporting.

3 Integration

This chapter gives necessary information for the integration of the MICROSAR vLinkGen into an application environment of an ECU.

3.1 Embedded Implementation

The delivery of the vLinkGen contains these source code files:

File Name	Description	Integration Tasks
vLinkGen_Lcfg.c	Generated file that contains initialization tables (see 2.4).	-
vLinkGen_Lcfg.h	Generated file that contains type definitions and extern declarations of the initialization tables.	-
vLinkGen_Cfg.h	Generated file that contains pre-compile time macros.	-

Table 3-1 Implementation files

In addition to the source code files mentioned above, vLinkGen also generates one or more compiler specific linker scripts or snippets.

The following Table 3-2 shows the different files that can be generated. Which of these variants is generated depends on the **File Generation** parameter, which is described in chapter 2.1.

File Name	Description	Integration Tasks
vLinkGen_Template.<ext>	Compiler specific linker script.	May have to be preprocessed and then passed to linker via specific option.
vLinkGen_Template_<Variant>.<ext>	Compiler specific linker script for a dedicated variant. Where <Variant> is the name of the variant.	Must be passed to linker via specific option.
vLinkGen_Inc_<Name>.<ext>	Compiler specific linker script snippet. Where <Name> can be the name of a module, variant or Logical Group.	Must be included within another linker script. Cannot be used stand-alone.

Table 3-2 Generated linker scripts and snippets

Where <ext> is the compiler specific file extension for linker scripts.



Note
The linker scripts are generated in the **“Template Files”** (Source) folder of the DaVinci Configurator Project.

These files are to be seen as templates and can be changed manually at any time.

Please note that the files will be overwritten any time the vLinkGen configuration changes. Therefore, if the files are to be modified manually, they should be copied to another location.

3.2 Compiler Restrictions

The following chapter indicates which features cannot be used at all or only with restrictions on certain compilers.

Parameters which are set to read-only in the DaVinci Configurator are not supported for the selected compiler.

3.2.1 ARM

- The vLinkGen does not generate any labels (e.g. for Section Groups) for this compiler. However, the linker itself creates start and end labels for each Section Group. For details, please refer to chapter 3.4.

3.2.2 Green Hills

- The "DEFAULTS" area in the linker script is currently not supported.

3.2.3 Renesas

- Alignments are only generated for Linker Sections and are restricted to 16 bytes.
- The vLinkGen does not generate any labels (e.g. for Section Groups) for this compiler. However, the linker itself creates start and end labels for each Linker Section. For details, please refer to chapter 3.4.

3.2.4 Tasking

- The linker must be called with option `-Wl--user-provided-initialization-code`.
- If the linker is called with option `--section-name-with-symbol`, all configured Linker Sections might need a wildcard (*) configured in their **Filenames** (`vLinkGenLinker*SectionFiles`) parameter. Otherwise the Linker Sections might be linked to another location then expected.
- The functionality of the **KEEP** flag is only guaranteed from version v6.1r1 of the Tasking Compiler. It may work with earlier versions also, but this is not guaranteed. The KEEP flag can be configured for any vLinkGenLinkerSections via the **Compiler Specific Flags** (`vLinkGenLinker*SectionFlag`) parameter.

3.2.4.1 Workaround for TCVX-39968

The following chapter describes a workaround for the following Tasking linker issue:

"*TCVX-39968 When using GROUPing, the order of the ROM copies of initialized data is different to their RAM order*".

3.2.4.1.1 Error Description

The order of the variables of a RAM group differs in the created ROM copy. This can cause the variables to be initialized with incorrect values, which in turn can lead to unexpected behavior. The re-order is caused when using the wildcard operator (e.g. "`[.]data*`") and if variables are present where one name contains the other completely and from the beginning, for example:

> `.data.file.GlobalVar`

> `.data.file.GlobalVar`**`Ext`**

The following extract of a map file shall illustrate the issue. Where the order of two global variables in RAM (`Data_Default`) and ROM (`Data_Default_INIT_ROM`) is reversed. Where sections in brackets ("`[...]`") are always the ROM copies of initialized RAM sections created by the linker.



Extract of Tasking map file

```
...
********************************  Locate Result  ********************************
...
| mpe:dspr0 | Data_Default          | .data.file.GlobalVar      | 0x7000af20 | ... |
| mpe:dspr0 | Data_Default          | .data.file.GlobalVarExt   | 0x7000af2C | ... |
...
| mpe:flash | Data_Default_INIT_ROM | [.data.file.GlobalVarExt] | 0xa0066480 | ... |
| mpe:flash | Data_Default_INIT_ROM | [.data.file.GlobalVar]    | 0xa006648C | ... |
...
```

3.2.4.1.2 Preconditions

The issue only occurs if the following conditions are met:

- The used Tasking compiler version is v4.2.r2 or lower.
- The source code contains global variables which are not linked to a user defined linker section (e.g. via AUTOSAR MemMap mechanism)
  - Those variables will then automatically be linked into the `Data_Default` section group, respectively at the position of "`select "[.]data*"`".

3.2.4.1.3 Workaround

To work around this issue, all affected variables must be specified explicitly in the linker script.

1. Search for variables in the project map file that meet the above-mentioned preconditions and note them.

2. Create a new `vLinkGenLinkerSpecSection` for each variable with the **Short-Name** set to the name of the variable (e.g. `GlobalVar`).
   1. Set the **Value** (`vLinkGenLinkerSpecSectionValue`) of the section to the complete section name from the map file (e.g. `.data.file.GlobalVar`). Where `file` is the name of the object file in which the variable is defined.
3. Duplicate the `vLinkGenVarSectionGroup` **data** and remove all referenced **Sections** from the new one. The Section Group is a child of the `vLinkGenLogicalVarGroup` **Data_Default**. The name of the new Section Group could be **data_GlobalVars**.
   1. Ensure that **Init Policy** and **Init Stage** parameters have the same values as in the **data** Section Group.
4. Then add all created `vLinkGenLinkerSpecSections` as **Sections** to the new **data_GlobalVars** Section Group.
5. Finally, set the **Position** of the new **data_GlobalVars** Section Group to a value lower than the position of the **data** Section Group. This ensures that it is generated and linked before the **data** Section Group. Otherwise, the wildcard operator within **data** will also match the explicitly specified linker sections.

3.2.5 Texas Instruments

- The actual order of the Section Groups after linking may differ from the configured position (`vLinkGen*GroupPosition`). This is a restriction of the compiler and cannot be influenced by vLinkGen.

3.3 Output Generation Order

The following table provides an overview of the individual output types and their order in the generated linker script.

Output Type	Config Container	Sort Order
Memory Region Blocks	`vLinkGenMemoryRegionBlock`	All Memory Region Blocks within a Memory Region are sorted by their boundary (lower, upper) and then by position.

Logical Groups	`vLinkGenLogicalConstGroup`
	`vLinkGenLogicalSymbolGroup`
	`vLinkGenLogicalUserGroup`
	`vLinkGenLogicalVarGroup`
		All Logical Groups are sorted by their position and then by container name. Whereby the list of all Logical Groups, regardless of type, is sorted.

Section Groups	`vLinkGenConstSectionGroup`
	`vLinkGenUserSectionGroup`
	`vLinkGenVarSectionGroup`
		All Section Groups within a Logical Group are sorted by their position and then by container name.

Linker Sections	`vLinkGenLinkerCodeSection`
	`vLinkGenLinkerConstSection`
	`vLinkGenLinkerSpecSection`
	`vLinkGenLinkerStackSection`
	`vLinkGenLinkerVarSection`
		All Linker Sections within a Section Group are sorted by their container name.

Linker Symbols	`vLinkGenLinkerSymbol`	All Linker Symbols within a Logical Symbol Group are sorted by their configured name.

Table 3-3 Output generation order



Note
The order of the Section Groups in the generated linker script also depends on the position of their parent Logical Group. These are also sorted internally by their position and then by their name.

3.4 Label Generation

The following table shows an overview of the generated labels per compiler and which values they get after linking. The shown labels are either created by the respective compiler or added by vLinkGen.

The table is not complete. There may be other compiler specific labels. Please refer to the corresponding compiler manual for details.

Compiler	Label	Value
ARM	Image$$**<Name>**$$Base	Start address
	Image$$**<Name>**$$Limit	Start address + Size
Diab, Green Hills, GCC, Linaro, HighTec, Tasking, Texas Instruments	[_]_**<Name>**_START	Start address
	[_]_**<Name>**_END	Start address + Size - 1
	[_]_**<Name>**_LIMIT	Start address + Size
IAR	[_]_**<Name>**_START (**<Name>**$$Base)	Start address
	[_]_**<Name>**_LIMIT (**<Name>**$$Limit)	Start address + Size
Renesas	__S_**<Name>**	Start address
	__E_**<Name>**	Start address + Size

Table 3-4 Label Generation

Note
Where **<Name>** can be the name of a Memory Region Block, Section Group or Linker Section.

The LIMIT labels are used for the initialization tables.

The END labels are usually used for configuring end addresses of MPU regions. Because MPU region end addresses mostly need to be set up with the last writable address.

In addition, the vLinkGen generates also labels for all Logical Groups like the following:

```
[_]_<LogicalGroupName>_ALL_[START|END|LIMIT]
```

Those labels wrap all Section Groups configured within one Logical Group. That means, the START label points to the _<SectionGroupName>_START label of the first Section Group. The END and LIMIT labels point to the respective labels of the last Section Group.

Caution
These Logical Group labels are to be used with care (e.g. for MPU configuration) as it is not guaranteed, that the linker links all Section Groups, respectively all Linker Sections of a Logical Group next to each other.

3.5 Special Section Generation

The following table shows how Linker Sections with special values (vLinkGenLinkerSpecSection) are generated in the compiler specific script language.

Where parts in brackets (“[…]”) are optional and only generated if configured. Strings in angle brackets (“<…>”) are replaced with the corresponding parameter values.

An example how to configure a section with special values can be found in chapter 5.2.1.3.

Compiler	Linker Script Syntax
ARM	`*(<Value>[, OVERALIGN <Alignment>])`
Diab	`.<Short-Name>[ KEEP][ (NOLOAD)][ ALIGN(<Alignment>)] : { <Value> }`
GCC, Green Hills, HighTec, Texas Instruments	`[. = ALIGN(<Alignment>);] *(<Value>)`
IAR	`Block <Short-Name>_BLOCK { section <Value> },`
Renesas	`<Value>[, &]`
Tasking	`select "<Value>";`

Table 3-5 Special Section Generation

3.6 Migration of Existing Projects

For existing DaVinci Configurator projects containing a configuration of the vLinkGen in version 1.07.x, the version 2.xx.xx provides the possibility to migrate this configuration to the new BSWMD structure. This means that the first time this project is opened with the new vLinkGen version, the configuration will be automatically adjusted to fit the changed BSWMD and only minimal manual adjustments are necessary.



Note
The preferred way to migrate an existing project would be to add vLinkGen completely new to the project. This means first removing the module from the project settings and then adding it again.

However, this way should only be used if no manual changes have been made to the vLinkGen configuration (e.g. adding custom linker sections). Otherwise they will be deleted with the module. In this case, the migration should be performed using the following steps.

The following conditions must be observed before migration:

- The version of vLinkGen must have been at least 1.07.xx before.
- The vBaseEnv must be upgraded to version 2.xx.xx.
- If the MICROSAR OS is included in the project, it must also be upgraded to version 2.45.00 or higher.
- The existing configuration of vLinkGen must not have any errors.
  With warnings it may work, but this is not guaranteed.

To perform the migration, the following steps must be performed:

1. The vLinkGen, vBaseEnv and possibly OS files in the SIP must be exchanged with the new files before the migration.
2. The **<SipId>** in the project file of the DaVinci Configurator must be changed. This should be done automatically for update deliveries.
3. Open the project with the DaVinci Configurator.
4. If the changes are recognized correctly, a pop-up window named **SIP Mismatch** will appear.
5. Select the option **Migrate the project to this SIP and continue** and start migration with **OK**. This may take some time.
6. After the migration some error messages may appear in the **Validation** window. These can be easily fixed with the **Solve All** button. This may take some time again, depending on the size of the configuration.
7. Now remaining error messages of the vLinkGen must be fixed manually, because it can't find a suitable solution at this point. The displayed error messages should be the following:
   - **Cfg00022 - Missing parameter value**

     The Memory Region Block reference must be selected for each Logical Group.
   - **VLINKGEN01300 - Invalid init core ID configuration**

     These errors can all be solved simultaneously via the Solve all with ... option.
   - **VLINKGEN03260 - Invalid VAR section group init configuration**

     The Memory Region Block (in ROM) in which the initialization data of variables should be located must be selected here.
8. The configuration is now successfully migrated, and the new linker script can be generated.



Note
Only the configuration of the vLinkGen module is changed during the migration. All other modules are not affected by the changes.

4 API Description

4.1 Type Definitions

The types defined by the vLinkGen are described in this chapter.

4.1.1 vLinkGen_MemArea

This structure contains the information for a generic memory area to be initialized.

Struct Element Name	C-Type	Description	Value Range
Start	`uint32`	Start address of the memory area.	0..4294967295
End	`uint32`	End address of the memory area. Calculated from start address + size of the area.	0..4294967295
Core	`uint32`	The physical core ID to perform the initialization.	0..4294967295
Alignment	`uint32`	Optional ECC alignment of the Hardware Memory Area in bytes. If no alignment is set, the value is generated to zero.	0..4294967295

Table 4-1 vLinkGen_MemArea

4.1.2 vLinkGen_MemAreaSet

This structure contains the information for a set of generic memory areas.

Struct Element Name	C-Type	Description	Value Range
Num	`uint32`	Number of areas.	0..4294967295
Areas	`vLinkGen_MemArea`	Table of generic memory areas. See 4.1.1 type definition.	-

Table 4-2 vLinkGen_MemAreaSet

4.1.3 vLinkGen_RamMemArea

This structure contains the information for a RAM memory area to be initialized with concrete data from a ROM area.

Struct Element Name	C-Type	Description	Value Range
Start	`uint32`	Start address of the RAM memory area.	0..4294967295
End	`uint32`	End address of the RAM memory area. Calculated from start address + size of the RAM area.	0..4294967295
Romstart	`uint32`	Start address of the ROM memory area.	0..4294967295
Romend	`uint32`	End address of the ROM memory area. Calculated from start address + size of the ROM area.	0..4294967295
Core	`uint32`	The physical core ID to perform the initialization.	0..4294967295

Struct Element Name	C-Type	Description	Value Range
Alignment	`uint32`	Optional ECC alignment of the Hardware Memory Area in bytes. If no alignment is set, the value is generated to zero.	0..4294967295

Table 4-3 vLinkGen_RamMemArea

4.1.4 vLinkGen_RamMemAreaSet

This structure contains the information for a set of ROM to RAM copy memory areas.

Struct Element Name	C-Type	Description	Value Range
Num	`uint32`	Number of areas.	0..4294967295
Areas	`vLinkGen_RamMemArea`	Table of ROM to RAM copy memory areas. See 4.1.3 for type definition.	-

Table 4-4 vLinkGen_RamMemAreaSet

5 Configuration

MICROSAR vLinkGen is configured with Vectors DaVinci Configurator.

The descriptions of all configuration attributes are described with tool tips within the configuration tool.

They can easily be looked up during vLinkGen component configuration.

5.1 Configuration Variants

The vLinkGen supports the configuration variants

- `VARIANT-PRE-COMPILE`

The configuration classes of the vLinkGen parameters depend on the supported configuration variants. For their definitions please see the `vLinkGen_bswmd.arxml` file.

5.2 Configuration Hints

The following chapter gives some instructions how to configure the typical use cases in vLinkGen.

5.2.1 Linking of Linker Sections

The descriptions in this chapter assume that a C symbol (no matter if function, constant or variable) is assigned to a specific section in the source code via the AUTOSAR MemMap mechanism or manually (e.g. via `#pragmas`). The instructions refer only to the vLinkGen and the linking of sections in the memory layout.

To simplify the following descriptions, all specified BSWMD paths are based on the following root: `/MICROSAR/vLinkGen/vLinkGenMemLayout/`

5.2.1.1 Code or Constant Section

To place a constant section (with code or constant data) at a certain position in the memory layout, it must be configured in the Linker Sections (`vLinkGenLinkerSections`) as follows:

1. Create a new `vLinkGenLinkerCodeSection` or `vLinkGenLinkerConstSection` container.
2. Set the **Name** (`vLinkGenLinker*SectionName`) which will be the name of the section in the generated linker script. The **Short-Name** should correspond to the Name parameter, since it is also generated for some compilers.
3. Optionally, set an **Alignment** (`vLinkGenLinker*SectionAlignment`) for the section. This will align the start address of the section accordingly. Note that this is not supported by all compilers.
4. Optionally, mark the section for **Fast Addressing** (`vLinkGenLinker*SectionFast`).
5. Optionally, specify **Filenames** (`vLinkGenLinker*SectionFiles`) for the section. This could be the name of one or more files or archives (libraries) - with the files they contain. This instructs the linker to link only the section contents from these files.

Please refer to the parameter description in the configuration tool for details. A valid syntax would be for example:
“filea.o;lib1.a(file*.o,file_b.o);file_?.o”.

6. Optionally, set some **Compiler Specific Flags** (vLinkGenLinker*SectionFlag) for the section. If there are any available. These are compiler specific and their meaning must be taken from the corresponding manual.
7. After the section is configured, it must be assigned to a CONST Section Group (vLinkGenLogicalConstGroup/vLinkGenConstSectionGroup). This group determines where the section actually ends up in the memory layout.
8. To do this, the Section must be assigned to the **Sections** (vLinkGenConstSectionGroupRef) of a group.
9. The actual memory area in which the group is then placed is determined by the **Memory Region Block** (vLinkGenConstGroupRegion).
10. If this section is to be loaded into RAM at ECU startup, the Section Group must be configured accordingly. Details can be found in chapter 5.2.4.
11. After generating, the section will appear in the corresponding Section Group in the linker script.

5.2.1.2 Data or Variable Section

To place a section with variable data (to be initialized or not) at a certain position in the memory layout, it must be configured in the Linker Sections (vLinkGenLinkerSections) as follows:

1. Create a new vLinkGenLinkerVarSection container.
2. Set the **Name** (vLinkGenLinkerVarSectionName) which will be the name of the section in the generated linker script. The **Short-Name** should correspond to the Name parameter, since it is also generated for some compilers.
3. Optionally, set an **Alignment** (vLinkGenLinkerVarSectionAlignment) for the section. This will align the start address of the section accordingly. Note that this is not supported by all compilers.
4. Optionally, mark the section for **Fast Addressing** (vLinkGenLinkerVarSectionFast).
5. Optionally, specify **Filenames** (vLinkGenLinker*SectionFiles) for the section. This could be the name of one or more files or archives (libraries) - with the files they contain. This instructs the linker to link only the section contents from these files. Please refer to the parameter description in the configuration tool for details. A valid syntax would be for example:
“filea.o;lib1.a(file*.o,file_b.o);file_?.o”.
6. Optionally, set some **Compiler Specific Flags** (vLinkGenLinkerVarSectionFlag) for the section. If there are any available. These are compiler specific and their meaning must be taken from the corresponding manual.
7. After the section is configured, it must be assigned to a VAR Section Group (vLinkGenLogicalVarGroup/vLinkGenVarSectionGroup). This group determines where the section actually ends up in the memory layout.

8. To do this, the Section must be assigned to the **Sections** (`vLinkGenVarSectionGroupRef`) of a group.
9. The actual memory area in which the group is then placed is determined by the **Memory Region Block** (`vLinkGenVarGroupRegion`).
10. If this section is to be initialized with predefined values or zero at ECU startup, the Section Group must be configured accordingly. Details can be found in chapter 5.2.3.
11. After generating, the section will appear in the corresponding Section Group in the linker script.

5.2.1.3 Sections with Special Values

Normal sections are generated by the vLinkGen in the way the compiler expects (e.g. `*(*.<Section>)` for GCC). Sometimes it is necessary to bypass this syntax and generate a section the way it is configured. For this vLinkGen provides the Special Sections (`vLinkGenLinkerSpecSection`). Please refer to chapter 3.5 to see how they are generated for each compiler.



Note
Since the value is generated as it is configured, it must contain a valid syntax for the compiler used.

To generate such a Spec Section, it must be configured as follows:

1. Create a new `vLinkGenLinkerSections/vLinkGenLinkerSpecSection` container.
2. Set the **Value** (`vLinkGenLinkerSpecSectionValue`) to the expected value to be generated. Consider the compiler specific syntax.
3. Optionally, set an **Alignment** (`vLinkGenLinkerSpecSectionAlignment`) for the section. This will align the start address of the section accordingly. Note that this is not supported by all compilers.
4. Optionally, set some **Compiler Specific Flags** (`vLinkGenLinkerSpecSectionFlag`) for the section. If there are any available. These are compiler specific and their meaning must be taken from the corresponding manual.
5. Unlike the other section types, Spec Sections can be assigned to CONST or VAR Section Groups. The assignment is done in the same way via the corresponding **Sections** (`vLinkGen*SectionGroupRef`) parameter of a group.
6. After generating, the section will appear in the corresponding Section Group in the linker script.

5.2.2 Initialization of ECC Protected Memory

As described in chapter 2.4.1, vLinkGen provide the possibility to initialize ECC protected memory areas. This can be configured for individual Memory Region Blocks (`vLinkGenMemoryRegion/vLinkGenMemoryRegionBlock`) as follows:

1. Set the **Init Stage** (`vLinkGenMemoryRegionBlockInitStage`) to a value other than `NONE`. The different values determine the time of initialization during startup. See Table 2-4 for a list of all available stages.
2. Set the **Init Core ID** (`vLinkGenMemoryRegionBlockInitCore`) to the core which shall perform the initialization. The value must be the logical ID of the core.
3. vLinkGen then generates an entry for this block in the corresponding initialization table. For details about the generated initialization tables, please refer to chapter 2.4.

5.2.3 Initialization of VAR Section Group

As described in chapter 2.4.2, vLinkGen provides the possibility to initialize variables on ECU startup with either predefined values or zeros. To initialize a VAR Section Group (`vLinkGenLogicalVarGroup/vLinkGenVarSectionGroup`) it must be configured as follows:

1. Set the **Init Policy** (`vLinkGenVarSectionGroupInit`) to a value other than `NONE`.

INIT	Initialization values for variables are flashed to the ROM and then copied to RAM at ECU startup.
ZERO_INIT	Variables in RAM are written with zeros at ECU startup.
NONE	No initialization shall be performed for the Section Group.

Table 5-1 VAR Section Group Init Policies

1. Set the **Init Stage** (`vLinkGenVarSectionGroupInitStage`) to a value other than `NONE`. The different values determine the time of initialization during startup. See Table 2-4 for a list of all available stages.
2. Switch to the parent Logical Group.
3. If Init Policy is set to `INIT`, the **Memory Region Block – ROM Copy** (`vLinkGenVarGroupRomRegion`) has to be set. This is used to specify where the initialization values shall be located in ROM.
4. Set the **Init Core ID** (`vLinkGenVarGroupInitCore`) to the core which shall perform the initialization. The value must be the logical ID of the core.
5. The vLinkGen then generates entries in the corresponding initialization table. Refer to chapter 2.4 for details.



Note
The settings of the **Memory Region Block – ROM Copy** and the **Init Core ID** apply to all Section Groups within a Logical Group. If Section Groups need to be initialized by different cores or their initialization values shall be located somewhere else in ROM, the groups need to be moved to different Logical Groups.

5.2.4 Copy Code to RAM

As described in chapter 2.4.3, vLinkGen provides the possibility to copy code or constants from a ROM memory area to RAM. To copy a CONST Section Group (`vLinkGenLogicalConstGroup/vLinkGenConstSectionGroup`) from ROM to RAM it must be configured as follows:

1. Set the **Init Policy** (`vLinkGenConstSectionGroupInit`) to `COPY_TO_RAM`.
1. Set the **Init Stage** (`vLinkGenConstSectionGroupInitStage`) to a value other than `NONE`. The different values determine the time of initialization during startup. See Table 2-4 for a list of all available stages.
2. Switch to the parent Logical Group.
3. Set the **Memory Region Block - RAM Copy** (`vLinkGenConstGroupRamRegion`) to specify where the copy shall be located in RAM.
4. Set the **Init Core ID** (`vLinkGenConstGroupInitCore`) to the core which shall perform the initialization. The value must be the logical ID of the core.
5. The vLinkGen then generates entries in the corresponding initialization table. Refer to chapter 2.4 for details.



Note
The settings of the **Memory Region Block – RAM Copy** and the **Init Core ID** apply to all Section Groups within a Logical Group. If Section Groups need to be copy to different RAM locations, the groups need to be moved to different Logical Groups.

5.2.5 User Defined Linker Symbols

The vLinkGen also provides the possibility to configure global Linker Symbols compiler independently. That means it is possible to assign values to own symbols, and vLinkGen will generate these assignments in the appropriate syntax into the linker script. These symbols are configured within Logical Symbol Groups (`vLinkGenLogicalSymbolGroup`) as follows:

1. Create a new `vLinkGenLogicalSymbolGroup` container.
2. Set the **Position** (`vLinkGenSymbolGroupPosition`) of the group to specify their location within the linker script. Note that some compilers expect global symbols to be defined at a certain position in the linker script. In this case the position is simply ignored.
3. Optionally, it is possible to assign the group to one or more **Variants** (`vLinkGenSymbolGroupVariant`). More about Variants in chapter 2.2.
4. Create one or more `vLinkGenLinkerSymbol` containers within the Logical Group.
5. These symbols get a **Name** (`vLinkGenLinkerSymbolName`) and a **Value** (`vLinkGenLinkerSymbolValue`). With some compilers the values can also contain complex expressions.

6. After generating, the symbols appear in the corresponding syntax at the defined position in the linker script.



Caution
Some compilers do not support the definition of global Linker Symbols at all or only to a limited extent. This should be considered when using Linker Symbols. However, vLinkGen will issue warnings if a compiler does not support them or only to a limited extent.

5.2.6 Variant Handling

As described in chapter 2.2, vLinkGen provide the possibility to specify different variants of a linker script in one configuration. These variants can be added to the vLinkGen (`/MICROSAR/vLinkGen`) configuration as follows:

1. Add the optional `vLinkGenVariantHandling` container to the configuration.
2. The different variants can then be defined with the `vLinkGenVariant` sub-containers. Where each container reflects one variant. The name of the container is also the name of the variant.
3. The created variants can now be referenced by Logical Groups (`vLinkGenMemLayout/vLinkGenLogical*Group`) using the **Variants** (`vLinkGen*GroupVariant`) reference. This parameter allows the group to be assigned to one or more variants.
4. Finally, the **File Generation** shall be set accordingly. Please refer to chapter 2.1 for details.



Expert Knowledge
Only one variant is needed, but still a build process without the Logical Groups of this variant. Then a second variant can be created without being referenced by a Logical Group. Which then must only be defined in the corresponding build process.

5.2.7 Variant Specific Region Block Mappings

As described in chapter 2.1.1, vLinkGen provide the possibility to link certain CONST or VAR Logical Groups (respectively all their contained Section Groups) to different Hardware Memory Areas in different variants. These variant specific mappings can be added to the vLinkGen (`/MICROSAR/vLinkGen/vLinkGenMemLayout`) configuration as follows:

1. Add `vLinkGenVariantHandling` to the configuration as described in chapter 5.2.6. Create variants and reference them in Logical Groups (`vLinkGenLogical*Group`) as needed.

2. Navigate to the Logical Group (CONST or VAR) which Section Groups shall be linked to different Hardware Memory Areas and add the optional **Variant Handling** (`vLinkGen*GroupVariantHandling`) container.
3. Create one or more **Variant Specifics** (`vLinkGen*GroupVariantSpecifics`) container. One for each variant specific mapping. Where one Variant Specifics container can be used to change the mapping for multiple variants.
4. Optionally, set the **Overwriting Memory Region Block** (`vLinkGen*GroupSpecificRegion`) reference to specify where the Logical Group shall be linked in the specified variants. This value overwrites the **Default Memory Region Block** (`vLinkGenConstGroupRegion`) from the parent Logical Group.
5. Optionally, set the **Overwriting Memory Region Block – ROM/RAM Copy** (`vLinkGen*GroupSpecific*Region`) reference to specify where the Logical Group's copy data shall be linked to in the specified variants. This value overwrites the **Default Memory Region Block – ROM/RAM Copy** (`vLinkGen*Group*Region`) from the parent Logical Group.
6. Select the **Variants** (`vLinkGen*GroupSpecificVariant`) to which this specific mapping is to apply. Multiple selection is allowed.
7. Finally, the **File Generation** shall be set accordingly. Please refer to chapter 2.1 for details.



Note
The Overwriting Memory Region Block references described above are both optional. This means that for a variant, for example, only the ROM/RAM copy can be linked elsewhere.



Caution
Specified **Variants** (`vLinkGen*GroupSpecificVariant`) must also be added to the parent Logical Group Variants. Normally this is ensured by the vLinkGen Generator.

6 Glossary and Abbreviations

6.1 Glossary

Term	Description
BSWMD	The BSWMD is a formal notation of all information belonging to a certain BSW artifact (BSW module or BSW cluster) in addition to the implementation of that artifact.
Electronic Control Unit	Also known as ECU. Small embedded computer system consisting of at least one CPU and corresponding periphery which is placed in one housing.
Linker Script	Compiler specific description file which is used by the linker to know where to allocate sections in the target's memory. In compiler manuals often called “Linker Command File” or “Linker Description”.
Operating system	All management and executive programs of a computer and its resources comprising an available and clearly defined interface.

Table 6-1 Glossary

6.2 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basis Software
CAN	Controller Area Network protocol originally defined for use as a communication network for control applications in vehicles.
ECC	Error-Correcting Code memory
ECU	Electronic Control Unit
HIS	Hersteller Initiative Software
ID	Identifier (e.g. Identifier of a CAN message)
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
OS	Operating System
PLL	Phase-locked loop
RAM	Random Access Memory
ROM	Read Only Memory
SIP	Software Integration Package
vBRS	Vector Basic Runtime System

Table 6-2 Abbreviations

7 Contact

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